$$
\begin{aligned}
C_{\phi\,\partial^2}^{(21)\ (11)} \to \frac{1}{16\pi^2} \Big(& \frac{1}{4} m_1 \tilde{\mu} g_1^4 \mathrm{LF}_{2,2,0}[m_1, \tilde{\mu}] - \frac{3}{2} m_1 \tilde{\mu} g_1^4 \mathrm{LF}_{3,2,-1}[m_1, \tilde{\mu}] + m_1 \tilde{\mu} g_1^4 \mathrm{LF}_{3,3,-2}[m_1, \tilde{\mu}] + \\
& m_1 \tilde{\mu} g_1^4 \mathrm{LF}_{4,2,-2}[m_1, \tilde{\mu}] - \frac{1}{4} m_2 \tilde{\mu} g_2^4 \mathrm{LF}_{2,2,0}[m_2, \tilde{\mu}] - 4 m_2 \tilde{\mu} g_2^4 \mathrm{LF}_{3,2,-1}[m_2, \tilde{\mu}] + \\
& 3 m_2 \tilde{\mu} g_2^4 \mathrm{LF}_{3,3,-2}[m_2, \tilde{\mu}] + 3 m_2 \tilde{\mu} g_2^4 \mathrm{LF}_{4,2,-2}[m_2, \tilde{\mu}] - \frac{1}{2} \tilde{\mu} g_1^2 y_d^{pr} \overline{a_d}^{pr} \mathrm{LF}_{3,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] + \\
& \frac{1}{4} \tilde{\mu} g_1^2 y_d^{pr} \overline{a_d}^{pr} \mathrm{LF}_{3,2,-1}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] + \frac{1}{2} \tilde{\mu} g_1^2 y_d^{pr} \overline{a_d}^{pr} \mathrm{LF}_{4,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] - \\
& \frac{1}{2} \tilde{\mu} g_1^2 y_e^{pr} \overline{a_e}^{pr} \mathrm{LF}_{3,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] + \frac{1}{4} \tilde{\mu} g_1^2 y_e^{pr} \overline{a_e}^{pr} \mathrm{LF}_{3,2,-1}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] + \\
& \frac{1}{2} \tilde{\mu} g_1^2 y_e^{pr} \overline{a_e}^{pr} \mathrm{LF}_{4,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] + \frac{1}{8} \tilde{\mu} y_e^{pr} \overline{a_e}^{pr} \left(g_1^2 - g_2^2\right) \mathrm{LF}_{4,1,-1}\left[m_{\tilde{l}}^p, m_{\tilde{e}}^r\right] + \\
& \frac{1}{2} \tilde{\mu} g_2^2 y_e^{pr} \overline{a_e}^{pr} \mathrm{LF}_{5,1,-2}\left[m_{\tilde{l}}^p, m_{\tilde{e}}^r\right] - \frac{1}{8} \tilde{\mu} y_d^{pr} \overline{a_d}^{pr} \left(g_1^2 + 3 g_2^2\right) \mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r\right] + \\
& \frac{3}{2} \tilde{\mu} g_2^2 y_d^{pr} \overline{a_d}^{pr} \mathrm{LF}_{5,1,-2}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r\right] - \frac{3}{4} \tilde{\mu} g_2^2 y_u^{pr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,0}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] - \\
& \frac{1}{4} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} \left(g_1^2 + 6 g_2^2\right) \mathrm{LF}_{3,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] + \frac{1}{8} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} \left(g_1^2 + 9 g_2^2\right) \mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] + \\
& \frac{1}{4} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} \left(g_1^2 + 6 g_2^2\right) \mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] + \frac{3}{4} \tilde{\mu} g_2^2 y_u^{pr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,0}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p\right] + \\
& \frac{3}{4} \tilde{\mu} g_2^2 y_u^{pr} \overline{a_u}^{pr} \mathrm{LF}_{3,2,-1}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p\right] + \frac{1}{4} \tilde{\mu} y_u^{pr} \overline{a_u}^{pr} \left(2 g_1^2 - 9 g_2^2\right) \mathrm{LF}_{4,1,-1}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p\right] + \\
& \frac{3}{2} \tilde{\mu} g_2^2 y_u^{pr} \overline{a_u}^{pr} \mathrm{LF}_{5,1,-2}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p\right] - \frac{3}{8} m_1 \tilde{\mu} g_1^4 \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_1] - \frac{3}{8} m_1 \tilde{\mu} g_1^2 g_2^2 \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1] + \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^4 \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{2} m_1 \tilde{\mu} g_1^2 g_2^2 \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1] + m_2 \tilde{\mu} g_2^4 \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_2] + \\
& \frac{5}{8} m_2 \tilde{\mu} g_2^4 \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_2] - \frac{21}{8} m_2 \tilde{\mu} g_2^4 \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2] - \frac{3}{4} m_2 \tilde{\mu} g_2^4 \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \\
& \frac{3}{2} m_2 \tilde{\mu} g_2^4 \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2] + \frac{1}{2} \tilde{\mu} g_1^2 g_2^2 (7 m_1 + 5 m_2) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] - \\
& 2 \tilde{\mu} g_1^2 g_2^2 (m_1 + m_2) \mathrm{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] - \frac{9}{4} \tilde{\mu} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^r\right] + \\
& \frac{3}{4} \tilde{\mu} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r\right] - \\
& \frac{3}{4} \tilde{\mu} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r\right] - \frac{3}{8} \tilde{\mu} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \\
& \quad \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{q}}^s\right] + \frac{3}{4} \tilde{\mu} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] - \\
& \frac{3}{4} \tilde{\mu} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] - \frac{3}{8} \tilde{\mu} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \\
& \quad \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{q}}^p\right] - \frac{3}{2} \tilde{\mu} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{u}}^r, m_{\tilde{u}}^t, m_{\tilde{q}}^p\right] - \\
& m_1 \tilde{\mu} g_1^2 g_2^2 \mathrm{LF}_{2,1,1,0}[\tilde{\mu}, m_1, m_2] + \frac{1}{4} \tilde{\mu} g_1^2 g_2^2 (5 m_1 + m_2) \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] - \\
& \frac{1}{2} \tilde{\mu} g_1^2 g_2^2 (m_1 + m_2) \mathrm{LF}_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] - 2 \tilde{\mu} g_1^2 g_2^2 (m_1 + m_2) \mathrm{LF}_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] - \\
& \frac{3}{2} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] + \\
& \frac{3}{2} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] + \\
& \frac{3}{4} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{d}}^t, m_{\tilde{q}}^s\right] + \\
& \frac{3}{4} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^s, m_{\tilde{d}}^t, m_{\tilde{q}}^p\right] - \\
& \frac{3}{2} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] + \\
& \frac{3}{2} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] + \\
& \frac{3}{4} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{q}}^s\right] + \\
& \frac{3}{4} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^t, m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] - \\
& \frac{1}{2} \tilde{\mu}^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^p, m_{\tilde{l}}^s\right] + \\
& \frac{1}{2} \tilde{\mu}^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^p, m_{\tilde{l}}^s\right] + \\
& \frac{1}{4} \tilde{\mu}^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p, m_{\tilde{e}}^t, m_{\tilde{l}}^s\right] + \\
& \frac{1}{4} \tilde{\mu}^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^s, m_{\tilde{e}}^t, m_{\tilde{l}}^p\right] - \\
& \frac{1}{2} \tilde{\mu}^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^p, m_{\tilde{l}}^s\right] + \\
& \frac{1}{2} \tilde{\mu}^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^p, m_{\tilde{l}}^s\right] + \\
& \frac{1}{4} \tilde{\mu}^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^t, m_{\tilde{l}}^p, m_{\tilde{e}}^r, m_{\tilde{l}}^s\right] + \\
& \frac{1}{4} \tilde{\mu}^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^t, m_{\tilde{l}}^s, m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] + \\
& \frac{1}{4} \tilde{\mu}^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{l}}^p, m_{\tilde{l}}^s, m_{\tilde{e}}^r, m_{\tilde{e}}^t\right] + \\
& \frac{3}{4} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{d}}^t\right] - \\
& \frac{9}{4} \tilde{\mu} y_d^{sr} \overline{a_d}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{u}}^t\right] - \\
& \frac{9}{4} \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^t, m_{\tilde{u}}^r\right] + \\
& \frac{3}{4} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] - \\
& \frac{3}{4} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] - \\
& \frac{3}{8} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{q}}^s, m_{\tilde{u}}^t\right] - \\
& \frac{3}{8} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^t, m_{\tilde{q}}^s, m_{\tilde{u}}^r\right] + \\
& \frac{3}{4} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] - \\
& \frac{3}{4} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] - \\
& \frac{3}{8} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{u}}^t\right] - \\
& \frac{3}{8} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] - \\
& \frac{3}{2} \tilde{\mu} y_u^{sr} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{u}}^r, m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] \Big)
\end{aligned}
$$